STA 243: Introduction

Spring 2026

Welcome to STA 243!

- Syllabus has been uploaded to Canvas.
- Will post these slides later.
- Let's go over the class logistics.

Any questions?

Introduction to Computational Statistics

- **Computational Statistics is the new Mathematical Statistics!**
- This course is not a cookbook to implement some numerical methods in a programming language.
- The main purpose of this course is to understand the theoretical underpinning of designing algorithms for statistical problems.

Statistical Algorithms

- Theory of computation is a rich field in computer science. Why study it in a statistics curriculum?
- Often theory of computation studies worst case problems, and the data sets are assumed to be deterministic.
- In statistics, data sets are invariably random objects drawn from some distribution.
- So, in statistics we care about typical/average behavior of an algorithm rather than worst case behavior.
- Moreover, as your data is a random object, your algorithm is also a random function!
- Hence we need to start thinking about statistical algorithms as **Estimators**

Thought Experiment

- Suppose you are performing the simple task of adding two-digit numbers.
- But you are adding 1 billion numbers.
- Without any further information, you invariably have to perform the addition operation 1 billion times, each time you are given a set of 1 billion numbers.
- Suppose I tell you the number are drawn from a distribution with mean 22 and variance 4
- You now have more information about your data that could be leveraged to reduce the run time!

Thought Experiment (cont.)

- A simple estimator like (1 billion x 22) is already a good estimator of the true answer
- More fancy estimators can be constructed and their properties could be analyzed $\rightsquigarrow$ concentration of measure

Classical Statistics vs. Computational Statistics

- Illustrated through the MLE example: Let $\hat{\theta}_n = \arg\min_\theta \; -\sum_{i=1}^n \log p_\theta(x_i)$ be an estimator of a true parameter θ^*
- From your math stat courses, you know how to prove consistency: $\hat{\theta}_n \to \theta^*$ almost surely or in probability.
- Furthermore, you know how to show asymptotic normality: $\sqrt{n}(\hat{\theta}_n - \theta^*) \to N(0, V)$ for some matrix V (inverse Fisher information matrix).
- In practice, for most (real-world) problems, you won't be able to compute $\hat{\theta}_n$ exactly
- You will typically use an iterative algorithm to obtain $\tilde{\theta}_n$ which is close to $\hat{\theta}_n$

Classical Statistics vs. Computational Statistics (cont.)

- So $\tilde{\theta}_n$ is yet another estimator of θ^*.
- Whatever properties of MLE you studied in your math stat course, ideally you would like to show them for $\tilde{\theta}_n$
- A major part of computational statistics will be towards constructing estimators $\tilde{\theta}_n$ that are implementable/efficiently computable and have similar properties as that of the MLE estimator $\hat{\theta}_n$

Run time

- Formally: $T(n)$ is $O(f(n))$ iff for some constants c and n_0 we have $T(n) \leq cf(n)$ for all $n \geq n_0$
- Informally: $T(n) = O(f(n))$ means that $f(n)$ describes the upper bound for $T(n)$
- It tells you the kind of resource needs you can expect the algorithm to exhibit as your data gets bigger and bigger.
- From best (least resource requirements) to worst, the rankings are: $O(1), O(\log n), O(n \log n), O(n^2), O(n^3), O(2^n)$.

Run time plots

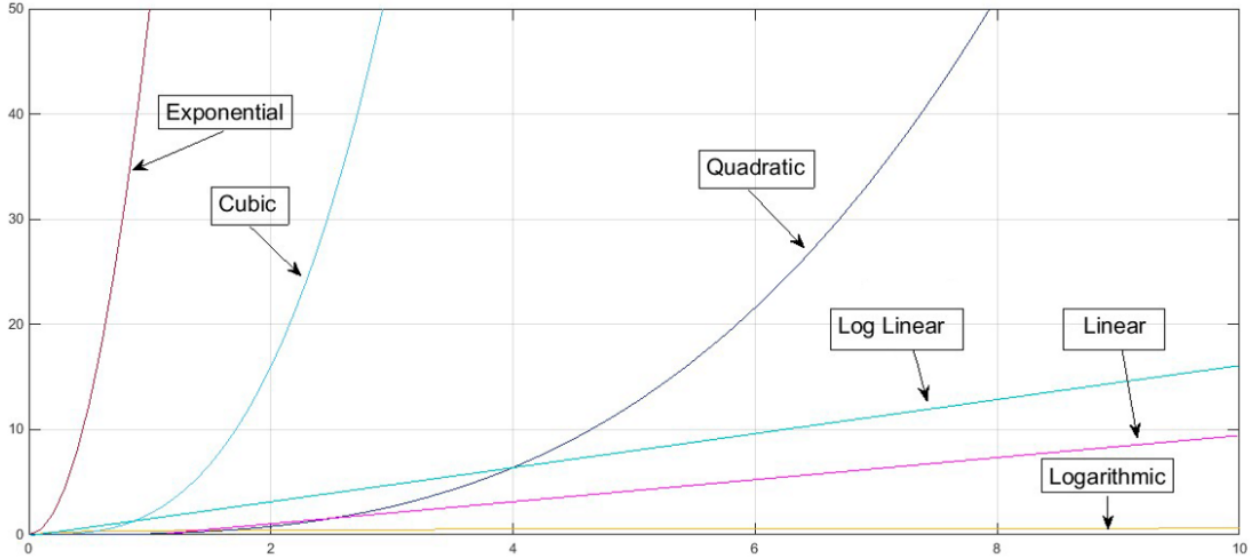

Figure: The growth rate of each curve determines the efficiency of algorithms (asymptotically).

Forms of Statistical Problems

- Every problem in Frequentist statistics can be formulated as an optimization problem. In this case, the solution could take one of the following forms:
  i. Closed-form solution available (typically computable in polynomial time)
  ii. Convex optimization (solvable in polynomial time even in the worst case)
  iii. Non-convex optimization (takes exponential time in worst case; exception: SVD)
  iv. Combinatorial optimization (takes exponential time in worst case)

  The last two classes can be solved in polynomial time in some statistical setting (which has led to some extremely interesting research questions in the past decade).

Forms of Statistical Problems (cont.)

- Every problem in Bayesian statistics can be formulated as a sampling problem.
  i. Sampling is typically harder than optimization
  ii. Several interesting relationship exists between sampling and optimization $\rightsquigarrow$ sampling as ∞-dimensional optimization in the space of probability measure.

These connections will often involve advanced tools from calculus of variations, stochastic analysis, PDE, probability etc.,

Forms of Statistical Problems (cont.)

Consider solving

$$x^* = \arg\min_x \ \{f(x) = (x-a)^2\}.$$

Through calculus you can set the derivative to zero and see that minimizer is $x^* = a$. But you can also think of this problem as calculating the expectation of the following density:

$$p(x) \propto e^{-\frac{f(x)}{\sigma^2}} = e^{-\frac{(x-a)^2}{\sigma^2}}$$

If you sample n point from the above density and compute the sample average, you are going to get a good estimate of $x^* = a$. Indeed, as $\sigma^2 \to 0$, it gets increasingly accurate as the density is concentrated around the expectation and you might need very few sample to get accurate estimates

Next: Introduction to Linear Algebra

- You are **highly encouraged** to brush up linear algebra
  i. `http://www.cs.cmu.edu/~zkolter/course/15-884/linalg-review.pdf`
  ii. `https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab`
- Topics required:
  i. Span, Basis vectors, Linear Independence
  ii. Range and Null space of matrices
  iii. Projections
  iv. Determinant and trace
  v. Quadratic forms, Eigenvalues and Eigenvectors
  vi. Matrix norms: Frobenius norm and operator norm
  vii. Matrix calculus (we will use this for optimization)

Prerequisite Test

1. State (as formally as you can) the three main modes of convergence of random variables.
2. State (as formally as you can) Chebyshev's inequality.
3. For a non-negative continuous random variable X, show that

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > t)\, dt$$

4. For a function $f : \mathbb{R}^d \to \mathbb{R}$, define its directional derivative. If in addition the function f is differentiable, define its gradient. What is the relationship between the directional derivative and gradient?
5. What are the four fundamental subspaces associated with a matrix $A \in \mathbb{R}^{m \times n}$? Define them formally and state (any) relationship between them.